No.

Date . .

1 decimal place

1. $X^2 + 5X + 1 = 0$

$a = 1$

$b = 5$

$c = 1$

$X_{1.2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$X_1 = -0.2$

$X_2 = -4.8$

b. $X^2 - 4X - 7 = 0$

$a = 1$

$b = -4$

$c = -7$

$X_{1.2} = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-7)}}{2(1)}$

$X_{1.2} = 2 + \sqrt{11}$

$X_1 = 5.3$

$X_2 = -1.3$

c. $X^2 - X - 11 = 0$

$a = 1$

$b = -1$

$c = -11$

$X_{1.2} = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-11)}}{2(1)}$

$X_{1.2} = \frac{1 \pm \sqrt{45}}{2}$

$X_1 = 3.9$

$X_2 = -2.9$

d. $4X^2 + 8X - 96$

simplified $\frac{4X^2 + 8X - 96}{X^2 + 2X - 24} \div 4$

$a = 4 = 1$

$b = 8 = 2$

$c = -96 = -24$

$X_{1.2} = \frac{-(2) \pm \sqrt{(2)^2 - 4(1)(-24)}}{2(1)}$

$X_{1.2} = \frac{-2 \pm \sqrt{100}}{2}$

$X_1 = 4$

$X_2 = -6$

e. $4X^2 + 20X + 25 = 0$

$a = 4$

$b = 20$

$c = 25$

$X_{1.2} = \frac{-(20) \pm \sqrt{(20)^2 - 4(4)(25)}}{2(4)}$

$X_{1.2} = -\frac{5}{2}$

$X_1 = -2.5$

$X_2 = -2.5$

f. $2X^2 - X - 9 = 0$

$a = 2$

$b = -1$

$c = -9$

$X_{1.2} = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(2)(-9)}}{2(2)}$

$X_{1.2} = \frac{1 \pm \sqrt{73}}{4}$

$X_1 = 2.4$

$X_2 = -1.9$

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No.

Date

2 decimal place

2 a. $10x^2 = -3x + 2$

zero product principle :

$10x^2 + 3x - 2 = 0$

$a = 10$

$b = 3$

$c = -2$

$$x_{1.2} = \frac{-(3) \pm \sqrt{(3)^2 - 4(10)(-2)}}{2(10)}$$

$$x_{1.2} = \frac{-3 \pm \sqrt{89}}{20}$$

$x_1 = 0.32$ $x_2 = -0.62$

C. $x(2x + 3) = 4 - 7x$

$2x$ $+3$

x $2x^2$ $3x$

Id $a(b + c) = ab + ac$

$x(2x + 3) = 2x^2 + 3x$

Zero product principle :

$x(2x + 3) = 4 - 7x$

$2x^2 + 3x = 4 - 7x$

$2x^2 + 3x + 7x - 4 = 0$

$2x^2 + 10x - 4 = 0$

$a = 2$

$b = 10$

$c = -4$

$$x_{1.2} = \frac{-(10) \pm \sqrt{(10)^2 - 4(2)(-4)}}{2(2)}$$

$$x_{1.2} = \frac{-5 \pm \sqrt{33}}{2}$$

$x_1 = 0.37$ $x_2 = -5.37$

b. $x(x - 1) = x + 5$

Id $a(b - c) = ab - ac$

$x(x - 1) = x^2 - x$

zero product principle :

$x(x - 1) = x + 5$

$x^2 - x = x + 5$

$x^2 - x - x - 5 = 0$

$x^2 - 2x - 5 = 0$

$a = 1$

$b = -2$

$c = -5$

$$x_{1.2} = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-5)}}{2(1)}$$

$$x_{1.2} = \frac{1 \pm \sqrt{6}}{}$$

$x_1 = 3.45$ $x_2 = -1.45$

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Date . .

2 decimal place

2. d $(x-2)(x-3) = 8$

	x	-3
x	x^2	$-3x$
-2	$-2x$	6

$-5x$

(d) distributive law:

$(x-2)(x-3) = 8$

$x^2 - 5x + 6 = 8$

$x^2 - 5x + 6 - 8 = 0$

$x^2 - 5x + (-2) = 0$

$x^2 - 5x - 2 = 0$

$a = 1$

$b = -5$

$c = -2$

$$x_{1,2} = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(-2)}}{2(1)}$$

$$x_{1,2} = \frac{5 \pm \sqrt{33}}{2}$$

$x_1 = 5.37$ $x_2 = -0.37$

2. e $3x + 5 = 5x^2 + 2(x-4)$

(i) $a(b-c) = ab - ac$

(i) $2(x-4) = 2x - 8$

reorder:

$3x + 5 = 5x^2 + 2x - 8$

$$\begin{array}{r} 5x^2 + 2x - 8 \\ -\quad\quad -3x - 5 = 0 \\ \hline 5x^2 - x - 13 = 0 \end{array}$$

$a = 5$

$b = -1$

$c = -13$

$$x_{1,2} = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(5)(-13)}}{2(5)}$$

$$x_{1,2} = \frac{1 \pm 3\sqrt{29}}{10}$$

$x_1 = 1.72$ $x_2 = -1.52$

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1 decimal place

g. $x^2 + 10x = 4$

zero product principle

$x^2 + 10x \boxed{-4} = 0$

$a = 1$

$b = 10$

$c = -4$

$x_{1.2} = \frac{-(10) \pm \sqrt{(10)^2 - 4(1)(-4)}}{2(1)}$

$x_{1.2} = \frac{-10 \pm \sqrt{116}}{2}$

$x_1 = 0.4$ $x_2 = -10.4$

h. $3x^2 + 7x + 4 = 1$

zero product principle :

$3x^2 + 7x + 4 \boxed{-1} = 0$

$3x^2 + 7x + \boxed{3}$

$a = 3$

$b = 7$

$c = 3$

$x_{1.2} = \frac{-(7) \pm \sqrt{(7)^2 - 4(3)(3)}}{2(3)}$

$x_1 = -0.6$ $x_{1.2} = \frac{-7 \pm \sqrt{13}}{6}$

$x_2 = -1.8$

i. $6x^2 - 5x = 8$

zero product principle

$6x^2 - 5x \boxed{-8} = 0$

$a = 6$

$b = -5$

$c = -8$

$x_{1.2} = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(-8)(6)}}{2(6)}$

$x_{1.2} = \frac{5 \pm \sqrt{217}}{12}$

$x_1 = 1.6$ $x_2 = -0.8$

j. $4x^2 + 2 = 7x$

zero product principle :

$4x^2 \boxed{-7x} + 2 = 0$

$a = 4$

$b = -7$

$c = 2$

$x_{1.2} = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(4)(2)}}{2(4)}$

$x_{1.2} = \frac{7 \pm \sqrt{17}}{8}$

$x_1 = 1.4$ $x_2 = 0.4$

No.

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3. A rectangle has a width of x m
a length of $(2x+5)$ m

"width and length are real number, so negative result will not be used."

If the area of rectangle $500\ m^2$,
find w : ?
L : ?
to the nearest metre.

width : 14.61 m
nearst metre
width = 15 m

Area = length × width
$500m^2 = x(2x+5)$

length = $2x + 5$ m
$= 2(14.61) + 5$ m
$= 34.22$ m
nearest metre
length = 34 m

(1d) distributive law
$a(b+c) = ab + ac$
$x(2x+5) = 2x^2 + 5x$

width : 15 m
length : 34 m

put it back to area formula
Area = $L \cdot w$
$500 = x(2x+5)$
$500 = 2x^2 + 5x$

zero product principle

$2x^2 \quad 5x \quad -500 = 0$

$a = 2$
$b = 5$
$c = -500$

$x_{1,2} = \frac{-(5) \pm \sqrt{(5)^2 - 4(2)(-500)}}{2(2)}$

$x_{1,2} = \frac{-5 \pm 5\sqrt{161}}{4}$

$x_1 = 14.61$
$x_2 = -17.11$

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4. A rectangular field

length : 20 m longer than width
length : (20 + x)

width : x

"width and length are real number so negative result will not be used"

width : 35 . 83 m

nearest metre

width : 36 m

The area of field is $2000\ m^2$,
find w : ?
L : ?
to the nearest metre.

length : w + 20
= 35.83 + 20
= 55 . 83

nearest1 metre

length : 56 m

area = w . L

$2000\ m^2$ = x (20 + x)

(1d) distributive law

a (b + c) = ab + ac

x (x + 20) = x^2 + 20 x

put it back to area formula :

area = w . l

2000 = x (x + 20)

2000 = x^2 + 20 x

zero product principle :

2000 = x^2 + 20 x

0 = x^2 + 20 x − 2000

a : 1
b : 20
c : -2000

$$x_{1.2} = \frac{-(20) \pm \sqrt{(20)^2 - 4(1)(-2000)}}{2(1)}$$

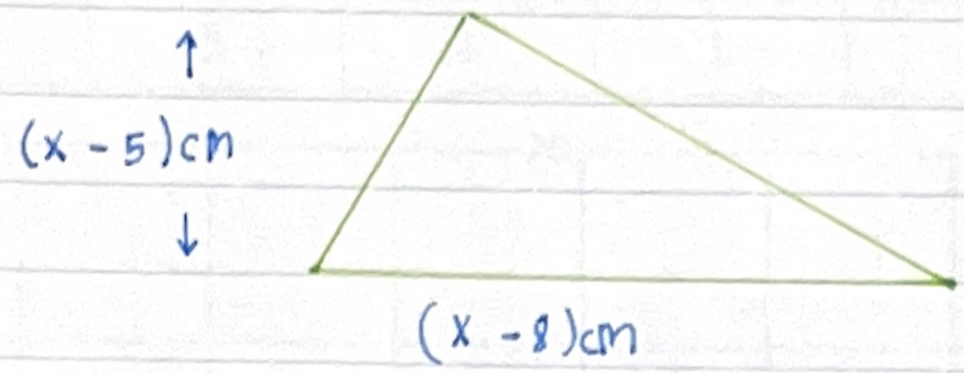
$$x_{1.2} = -10 \pm 10\sqrt{21}$$

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5.



Area = 30 cm²

(i) expand this

$(x - 5)(x - 8)$

$= x^2 - 8x - 5x + 40$

$= x^2 - 13x + 40$

	x	-8
x	x^2	$-8x$
-5	$-5x$	+40

(ii) Put in the area formula

$$\text{Area} = \frac{1}{2} b \cdot h$$

$$\text{Area} = \frac{b \cdot h}{2}$$

we can also use id: $(x - a)(x - b) = x^2 - (a + b)x + ab$

$(x - 5)(x - 8) =$

$x^2 - (5 + 8)x + (5)(8)$

$$\text{Area} = \frac{1}{2}(x - 5)(x - 8)$$

$$(2)\ 30 = (2)\frac{1}{2}(x^2 - 13x + 40)$$

No negative result !!

(iii) Reorder ar zero product principle

$60 = x^2 - 13x + 40$

$0 = x^2 - 13x + 40 - 60$

$0 = x^2 - 13x - 20$

$a = 1$

$b = -13$

$c = -20$

$$x_{1,2} = \frac{-(-13) \pm \sqrt{(-13)^2 - 4(1)(-20)}}{2(1)}$$

$$x_{1,2} = \frac{13 \pm \sqrt{249}}{2}$$

$x = 14.39$

base $= x - 8$

$= 14.39 - 8$

$= 6.39$ cm

nearest c.m

base : 6

height : $x - 5$

$= 14.39 - 5$

$= 9.39$ cm

nearest cm

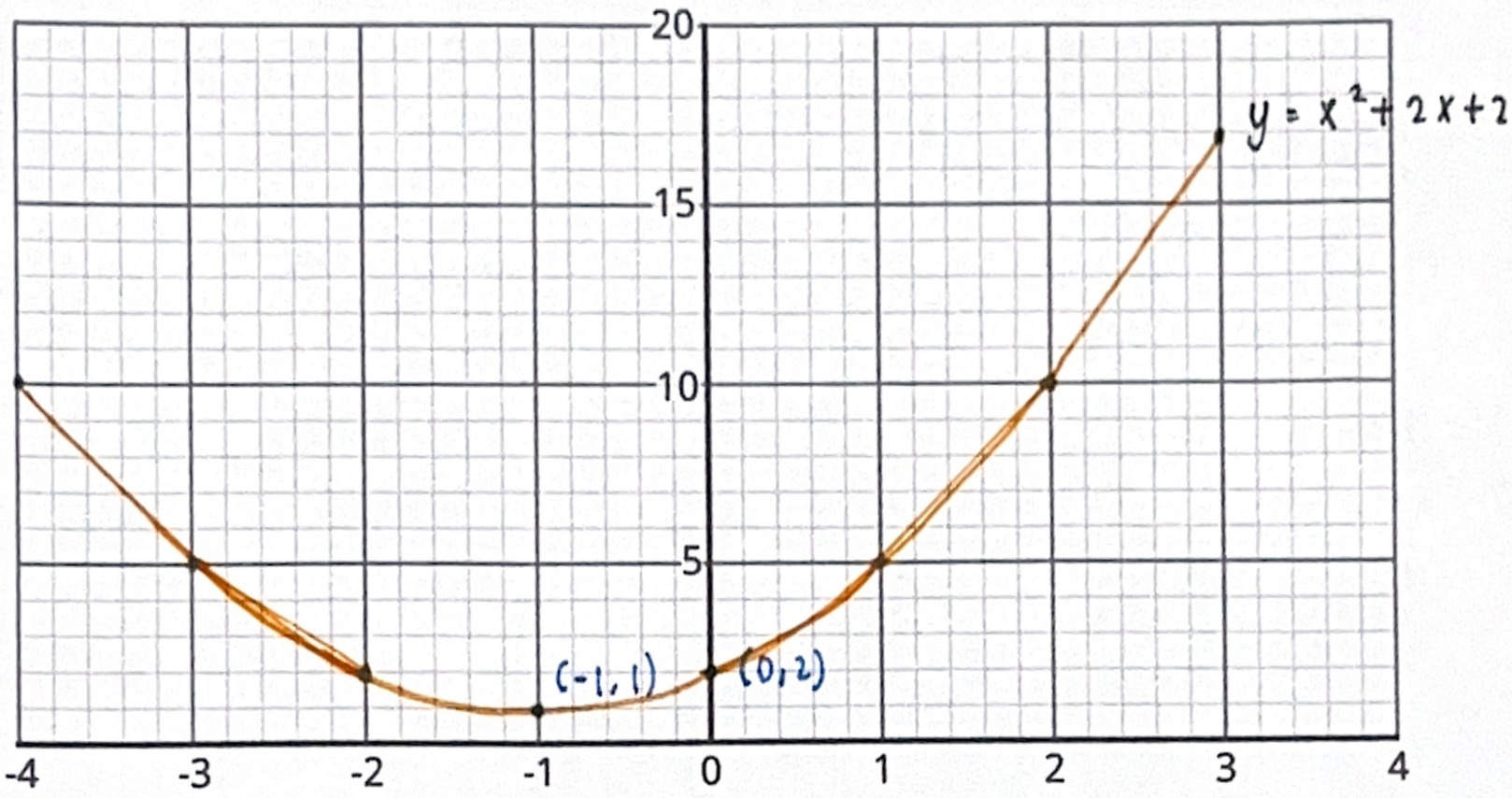
height : 9

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Your Turn

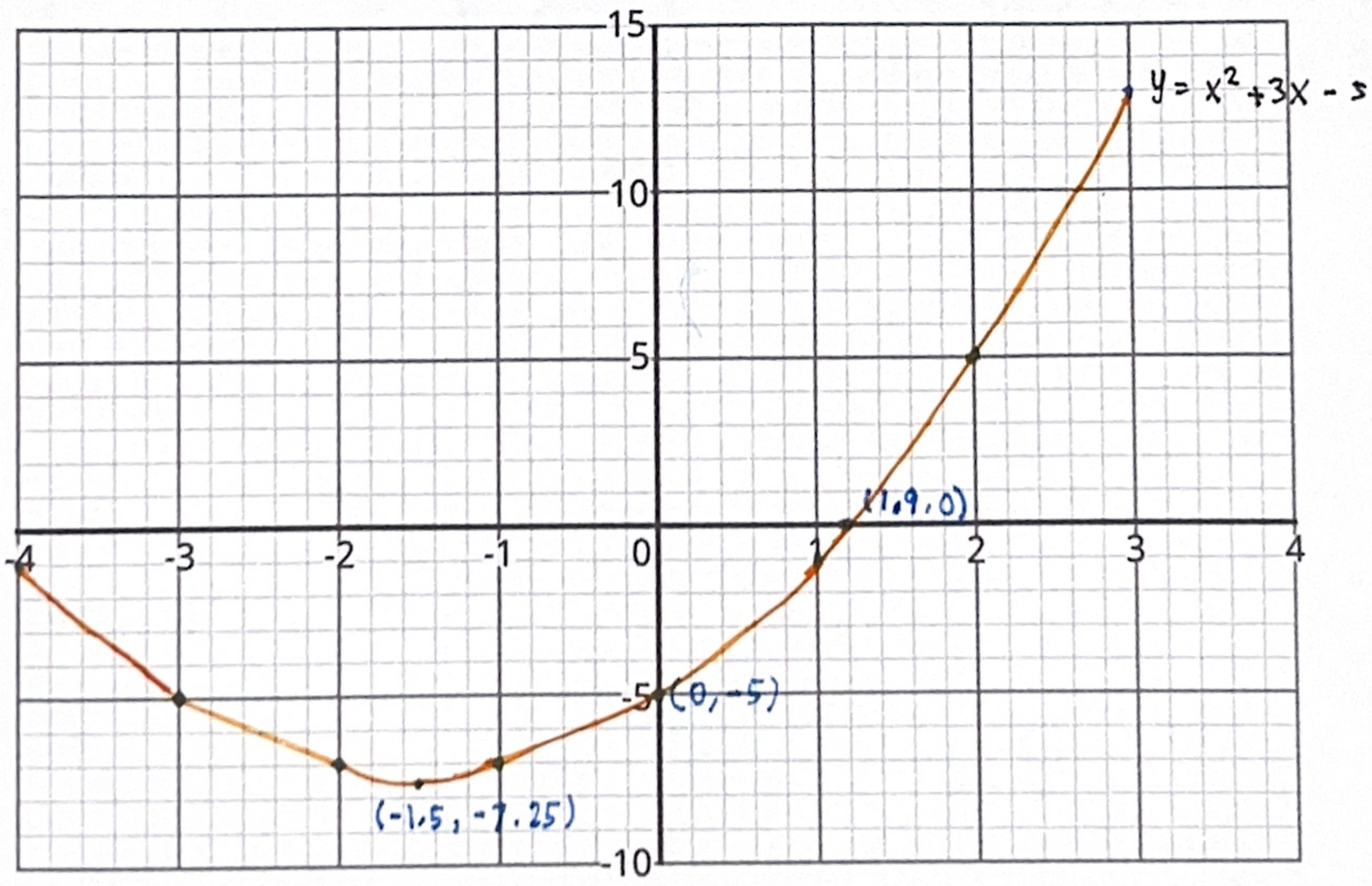
1. Draw the graph of $y = x^2 + 2x + 2$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y	5	2	1	2	5	10	17



2. Draw the graph of $y = x^2 + 3x - 5$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y	-5	-7	-7	-5	-1	5	13



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2. $y = x^2 + 3x - 5$

values of x : $-3 \leq x \leq 3$

x	-3	-2	-1
x^2	$(-3)(-3) = 9$	$(-2)(-2) = +4$	$(-1)(-1) = 1$
$3x$	$3(-3) = -9$	$3(-2) = -6$	$3(-1) = -3$
$x^2 + 3x$	$9 + (-9) = 0$	$4 - 6 = -2$	$1 + -3 = -2$
$x^2 + 3x - 5$	$0 - 5 = -5$	$-2 - 5 = -7$	$-2 - 5 = -7$
y	-5	-7	-7
coordinates	$-3, -5$	$-2, -7$	$-1, -7$

3. $y = 3x^2 - 2x - 6$

values of x : $-3 \leq x \leq 3$

x	-3	-2	-1
$3x^2$	$3(-3)(-3) = 27$	$3(-2)(-2) = 12$	$3(-1)(-1) = 3$
$2x$	$2(-3) = -6$	$2(-2) = -4$	$2(-1) = -2$
$3x^2 - 2x$	$27 - (-6) = 33$	$12 - (-4) = 16$	$3 - (-2) = 5$
$3x^2 - 2x - 6$	$33 - 6 = 27$	$16 - 6 = 10$	$5 - 6 = -1$
y	27	10	-1
coordinates	$-3, 27$	$-2, 10$	$-1, -1$

4. $y = -x^2 + x + 4$

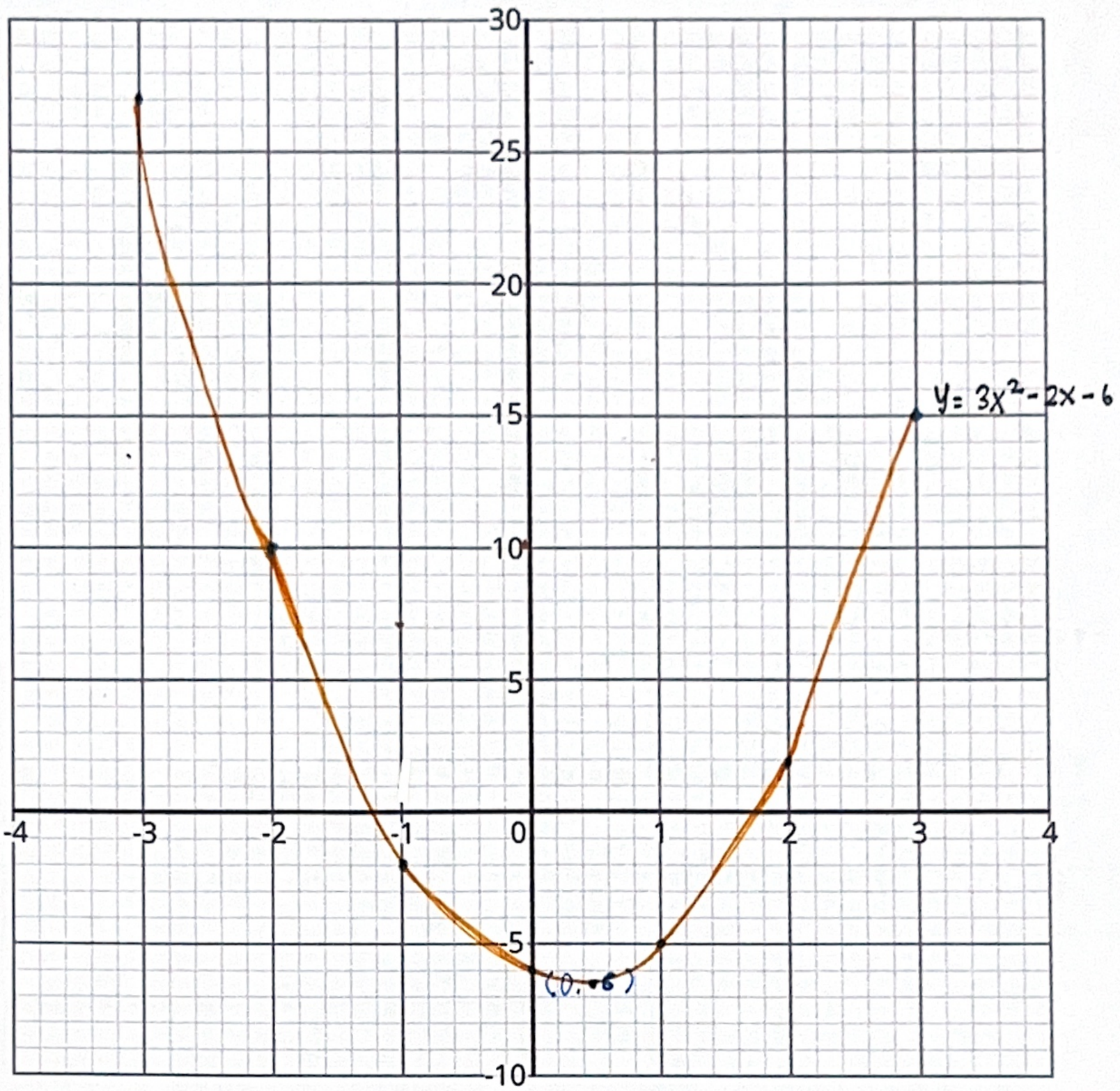
values of $x = -3 \leq x \leq 3$

x	-3	-2	-1
$-x^2$	$-(-3)(-3) = -9$	$-(-2)(-2) = -4$	$-(-1)(-1) = -1$
x	$(-3) = -3$	$(-2) = -2$	$(-1) = -1$
$-x^2 + x$	$-9 + (-3) = -12$	$-4 + -2 = -6$	$-1 + -1 = -2$
$-x^2 + x + 4$	$-12 + 4 = -8$	$-6 + 4 = -2$	$-2 + 4 = 2$
y	-8	-2	2
coordinates	$-3, -8$	$-2, -2$	$-1, 2$

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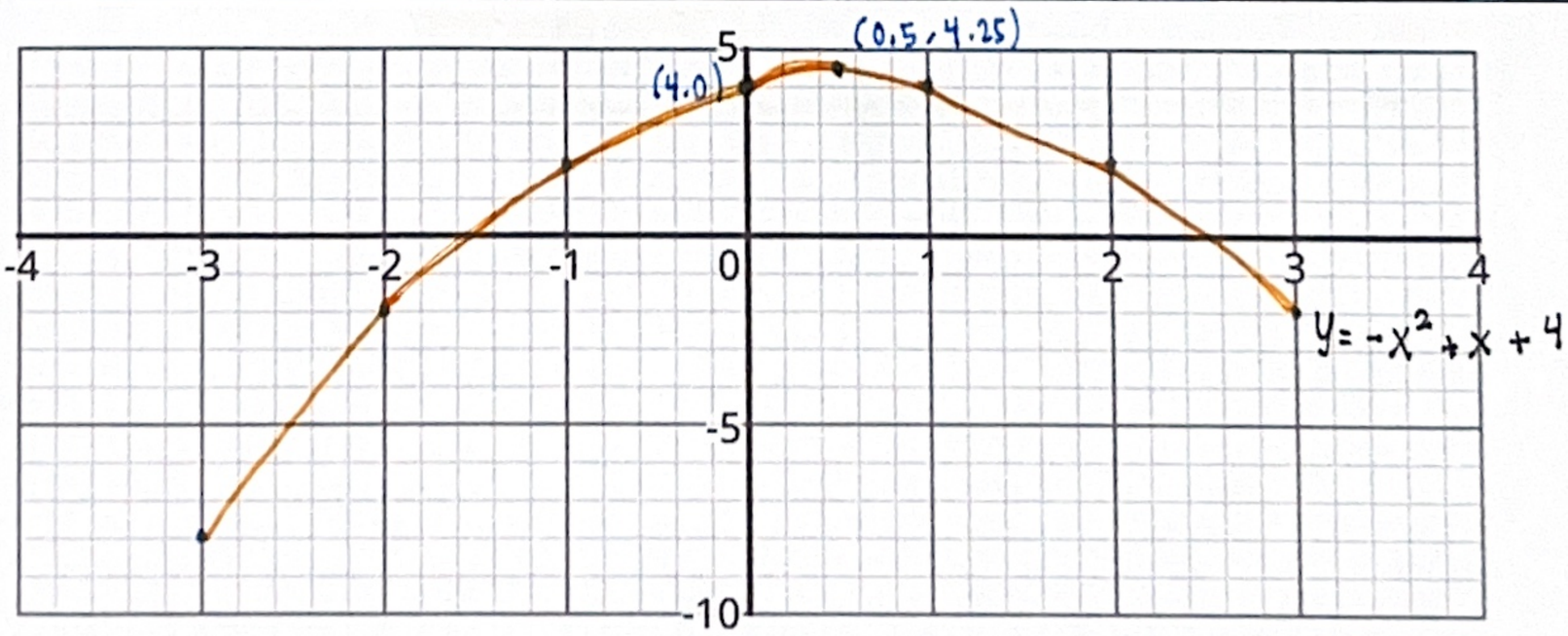
3. Draw the graph of $y = 3x^2 - 2x - 6$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y	27	10	-1	-6	-5	2	15



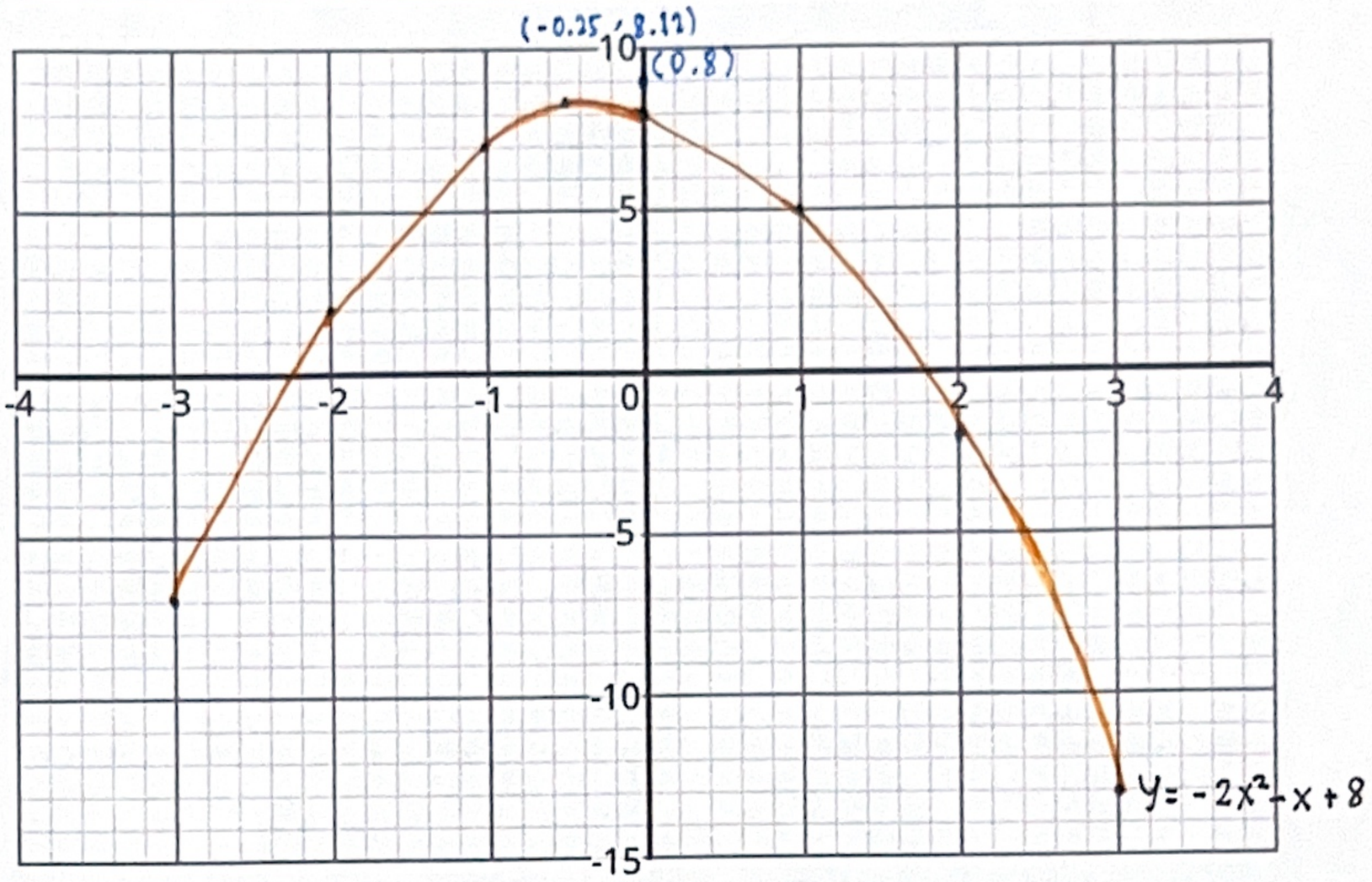
4. Draw the graph of $y = -x^2 + x + 4$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y	-8	-2	2	4	4	2	-2



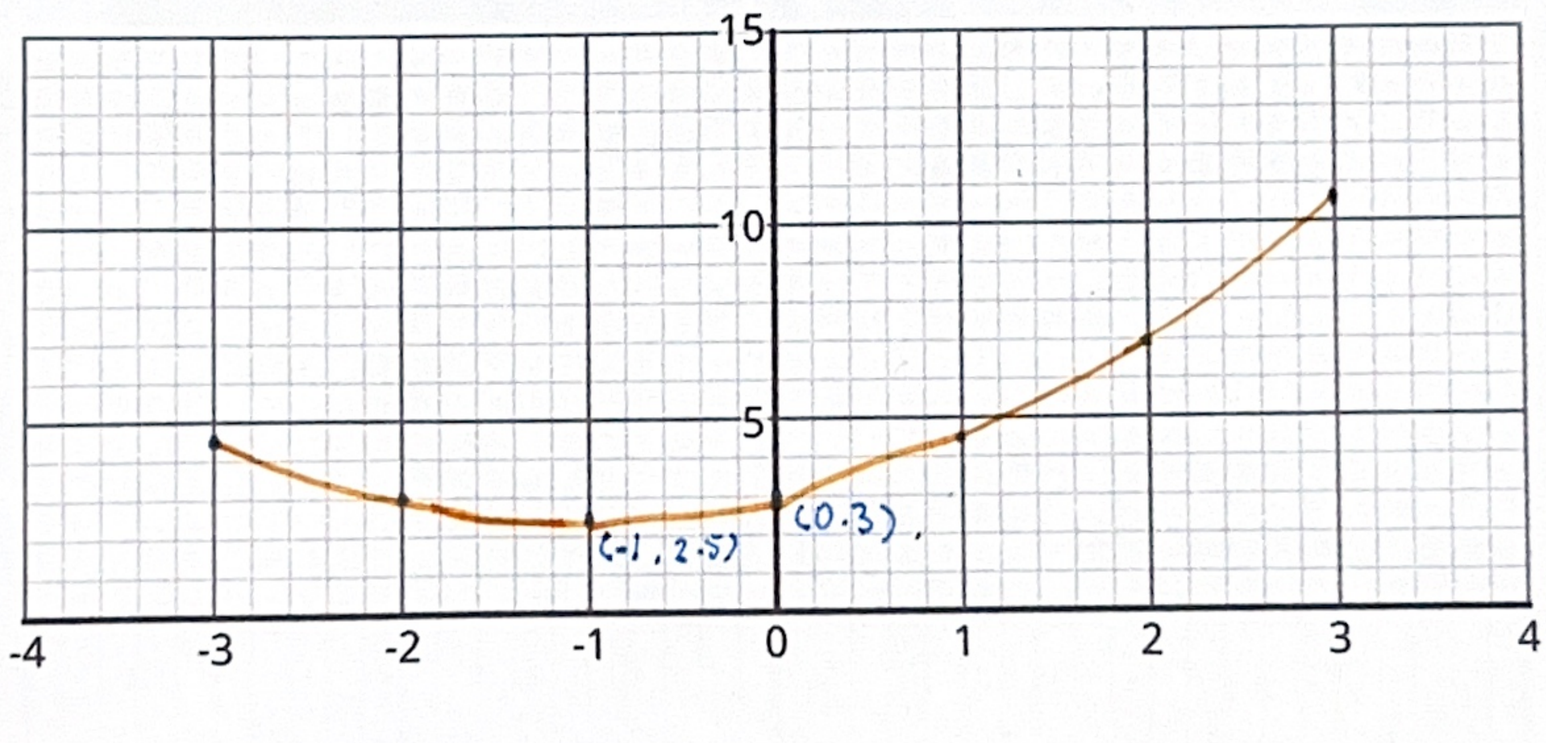
5. Draw the graph of $y = -2x^2 - x + 8$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y	-7	2	7	8	5	-2	-13



6. Draw the graph of $y = \frac{1}{2}x^2 + x + 3$ for the following values of x: $-3 \leq x \leq 3$

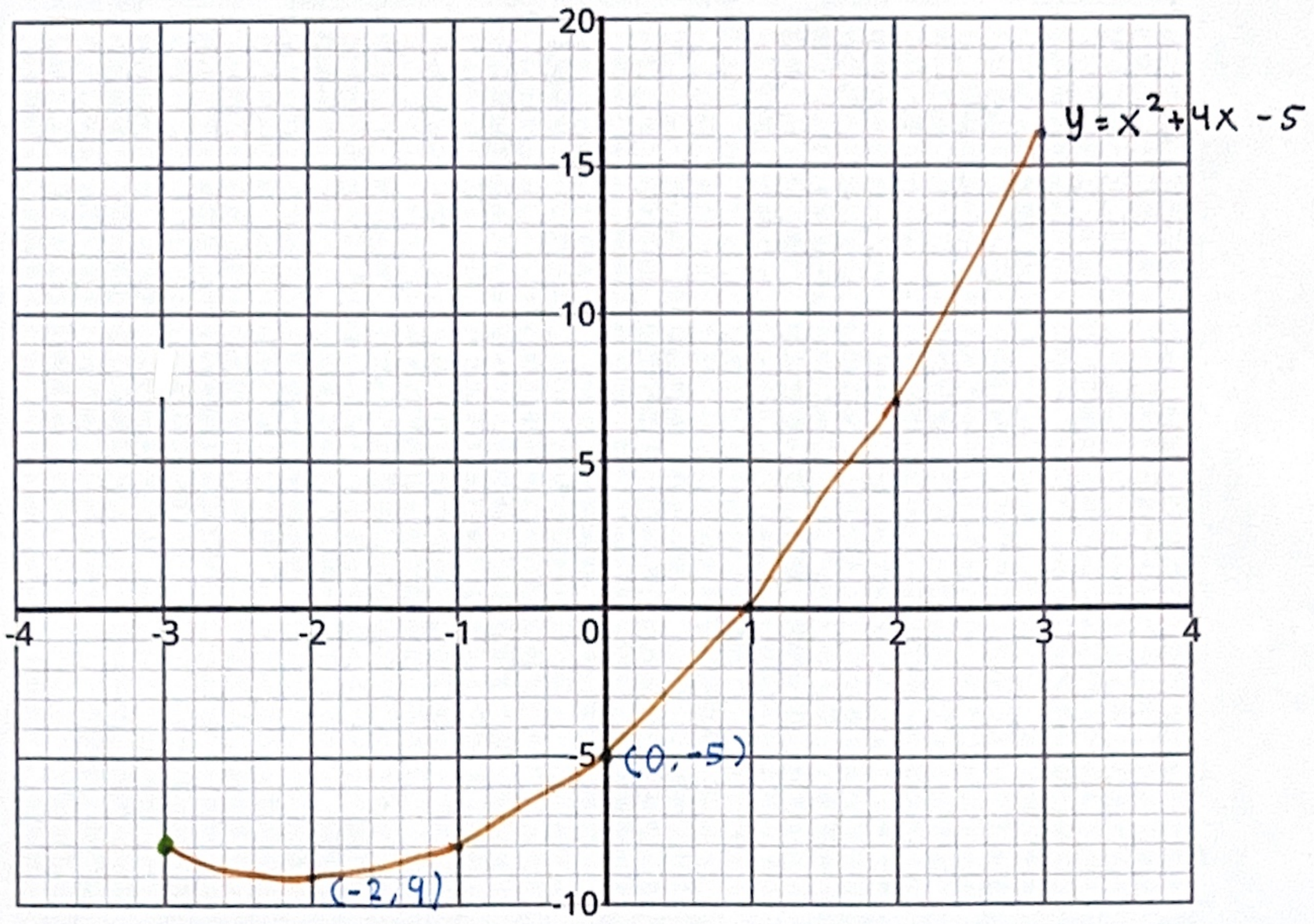
x	-3	-2	-1	0	1	2	3
y	4.5	3	2.5	3	4.5	7	10.5



Challenge

a. Draw the graph of $y = x^2 + 4x - 5$ for the following values of x: $-3 \leq x \leq 3$

x	-3	-2	-1	0	1	2	3
y	-8	-9	-8	-5	0	7	16



b. Hence, solve the equation $x^2 + 4x - 5 = 0$

Hint: Think about the word 'intercept'.

$y: x^2 + 4x - 5$

$x^2 + 4x - 5 = 0$

"intercept is factors"
intercept is solution

$(x - 1)(x + 5)$

zero product principle

$x - 1 = 0$ $\quad$ $x + 5 = 0$

$x = 1$ $\quad$ $x = -5$

two intercept
$(-5, 0)$ and $(1, 0)$

X	-3	-2	-1	0	1	2	3
Y	-8	-9	-8	-5	0	7	16
Coordinates	-3, -8	-2, -9	-1, -8	0, -5	1, 0	2, 7	3, 16